

### Foundation (Jalapeno)

- Q1.** Evaluate  $f(g(x))$  for  $f(x) = x^2$  and  $g(x) = 2x$ .  
(A)  $4x^2$   
(B)  $2x^2$   
(C)  $x^2$   
(D)  $x$   
(E)  $4x$
- Q2.** Find the domain of  $f(g(x))$  for  $f(x) = \frac{1}{x}$  and  $g(x) = x - 2$ .  
(A)  $(-\infty, 2)$   
(B)  $(-\infty, 2]$   
(C)  $[2, \infty)$   
(D)  $(-\infty, \infty)$   
(E)  $(-2, \infty)$
- Q3.** If  $f(x) = |x|$  and  $g(x) = x + 1$ , evaluate  $f(g(-2))$ .  
(A) 1  
(B) 3  
(C) -1  
(D) -3  
(E) 0
- Q4.** Determine the range of  $f(g(x))$  for  $f(x) = x^2$  and  $g(x) = 2x - 3$ .  
(A)  $[0, \infty)$   
(B)  $(-\infty, 0)$   
(C)  $[1, \infty)$   
(D)  $[9, \infty)$   
(E)  $(-\infty, -9]$
- Q5.** Find  $f(g(x))$  for  $f(x) = \sqrt{x}$  and  $g(x) = x^2 - 4$ .  
(A)  $\sqrt{x^2 - 4}$   
(B)  $\sqrt{x^2 + 4}$   
(C)  $x^2 - 4$   
(D)  $x^2 + 4$   
(E)  $x - 4$
- Q6.** Given  $f(x) = x^3$  and  $g(x) = \sqrt{x}$ , find  $f(g(4))$ .  
(A) 2  
(B) 8  
(C) 4  
(D) 16  
(E) 64
- Q7.** For  $f(x) = \frac{1}{x+1}$  and  $g(x) = 2x$ , evaluate  $f(g(2))$ .  
(A)  $\frac{1}{5}$   
(B)  $\frac{1}{3}$   
(C)  $\frac{1}{2}$   
(D)  $\frac{2}{3}$   
(E)  $\frac{1}{4}$
- Q8.** If  $f(x) = |x - 2|$  and  $g(x) = x + 1$ , evaluate  $g(f(-1))$ .  
(A) 2  
(B) 3  
(C) 1  
(D) 0  
(E) -1

#### Open Ended Questions (FRQ)

- Q9.** Consider the functions  $f(x) = x^2 - 2$  and  $g(x) = 2x + 1$ . Find  $f(g(x))$  and determine its domain and range.

- Q10.** Suppose  $f(x) = \frac{1}{x}$  and  $g(x) = \frac{1}{x+1}$ . Evaluate  $f(g(x))$  and interpret the result in the context of real-world applications.

#### Chef's Tips

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- Tip 1: Ensure students understand the concept of composite functions before the test.
- Tip 2: Emphasize the importance of checking the domain restrictions for composite functions.
- Tip 3: Provide real-world examples to help students interpret compositions in different contexts.